

The Method of Characteristics

First-Order Hyperbolic PDEs

Dr. Allan Struthers

Michigan Technological University

Motivation

- First-order PDEs describe *transport* and *propagation*
- Information travels along curves called *characteristics*
- The PDE reduces to ODEs along these curves
- Geometry explains existence, uniqueness, and breakdown

The Unit Linear Advection Equation

$$u_t + u_x = 0$$

- Simplest hyperbolic PDE
- Models transport to the right with unit speed
- Initial condition: $u(x, 0) = u_0(x)$

Characteristic Curves

We look for curves $x(t)$ along which u is constant.

$$\frac{d}{dt}u(x(t), t) = u_t + x'(t)u_x$$

Choose

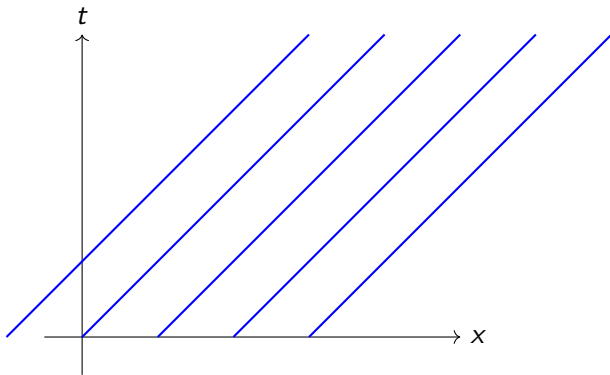
$$x'(t) = 1 \quad \Rightarrow \quad \frac{d}{dt}u = 0$$

Conclusion

u is constant along straight lines

$$x - t = \text{constant}$$

Characteristics: $u_t + u_x = 0$



All characteristics are parallel straight lines with slope 1.

Linear Advection with Speed c

$$u_t + cu_x = 0$$

- c is the transport speed
- Can be positive, negative, or zero
- Direction of propagation depends on sign of c

Characteristics for General c

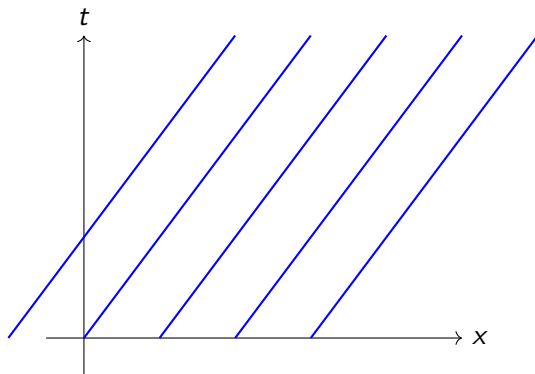
$$\frac{d}{dt}u = u_t + x'(t)u_x \quad \Rightarrow \quad x'(t) = c$$

$$x(t) = x_0 + ct$$

Key Point

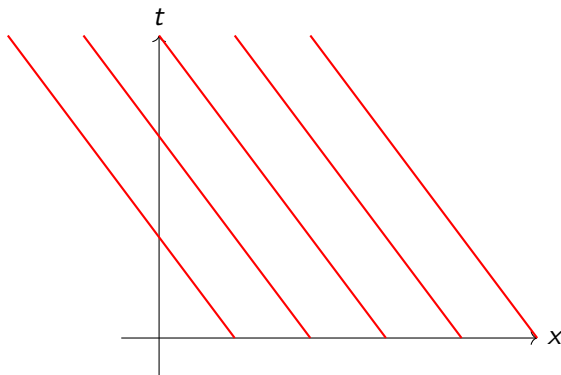
Initial data at x_0 moves rigidly with speed c .

Positive Speed $c > 0$



Information propagates to the right.

Negative Speed $c < 0$



Information propagates to the left.

Burgers' Equation

$$u_t + uu_x = 0$$

- Nonlinear transport equation
- Speed depends on the solution itself
- Prototype for shock formation

Compute

$$\frac{d}{dt}u(x(t), t) = u_t + x'(t)u_x$$

Choose

$$x'(t) = u$$

Then

$$\frac{d}{dt}u = 0$$

Key Consequence

$$u(x(t), t) = u_0(x_0)$$

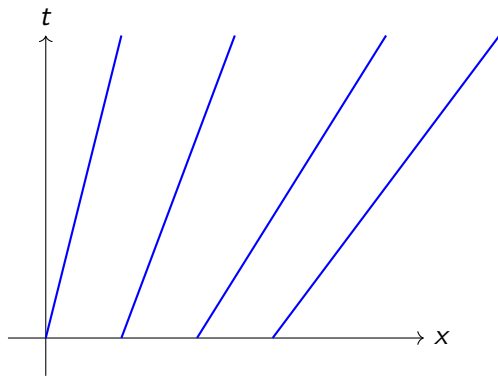
Parametric Form of Characteristics

For initial point x_0 ,

$$u = u_0(x_0)$$
$$x(t) = x_0 + u_0(x_0) t$$

- Each initial value defines its own speed
- Larger values move faster
- Smaller values lag behind

Characteristics for Burgers' Equation



Different slopes correspond to different initial values.

Characteristic Crossing

- Faster characteristics catch slower ones
- Crossing means multiple values of u
- Classical solution breaks down

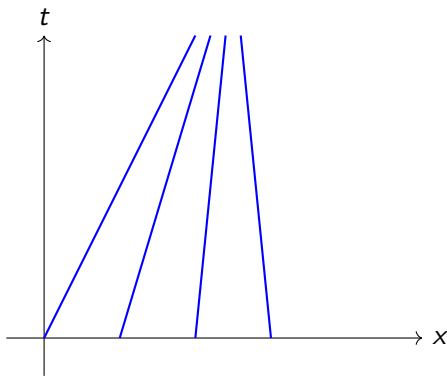
$$x = x_0 + u_0(x_0) t$$

When

$$\frac{\partial x}{\partial x_0} = 0$$

the mapping fails.

Shock Formation Picture



Characteristics intersect $\Rightarrow$ gradient blow-up.

The first shock occurs when

$$t_* = -\frac{1}{\min u'_0(x)}$$

- Requires decreasing initial data
- Steeper slopes $\Rightarrow$ earlier shock
- Purely geometric phenomenon

Summary

- Characteristics reduce PDEs to ODEs
- Linear advection: straight, parallel lines
- Sign of speed controls direction
- Burgers' equation: nonlinear geometry
- Crossing characteristics signal singularity formation